Assignment 3: Optical Flow, Video Retiming and Stabilisation

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Abstract

This report summarises work done for Assignment 3: implementations of a windowed Lucas–Kanade optical flow, a simple pyramid-based approach for frame interpolation, and global-translation video stabilisation. We apply the methods to the provided videos and present qualitative results and brief quantitative notes. All code used is included in the repository.

1 Introduction

This assignment implements classical computer-vision building blocks from scratch (NumPy/OpenCV): image gradients, windowed Lucas–Kanade (LK) flow, sparse-to-dense flow interpolation, simple warping for frame interpolation, and median-based translation stabilisation. A bonus vanishing-point detector based on line intersections is also included.

2 Methods

2.1 Gradients and Windowed Lucas–Kanade

We compute spatial gradients using Sobel filters and temporal gradient as frame difference. For sparse flow we solve the normal equations in local $n \times n$ windows: $A^T A\mathbf{v} = -A^T b$, where $A = [I_x \ I_y]$ and $b = I_t$. Windows with small eigenvalues are rejected to avoid ill-conditioning.

2.2 Sparse to Dense and Warping

Sparse flow vectors are converted to a dense field by low-resolution interpolation (resize small and upsample) as a lightweight approach for warping. We warp using pixel-remap (inverse mapping) and blend forward and backward halfway warps for interpolation.

2.3 Stabilisation

Global translation per frame is estimated as the median of sparse flow vectors across a regular grid. The cumulative camera path is smoothed with a short moving average and frames are inverse-warped by the smoothed path to produce the stabilized video.

2.4 Vanishing Point

Edges are detected (Canny), lines estimated using Probabilistic Hough Transform, and line intersections clustered (k-means with $k = 1$) to estimate a dominant vanishing point per frame; results are aggregated across frames.

3 Experiments and Results

We ran the implemented scripts on the provided videos. Representative outputs are included below.

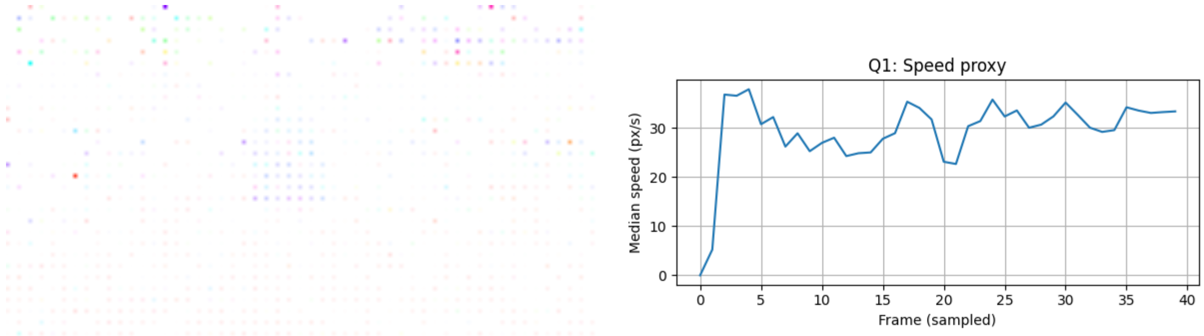

Figure 1: (Left) Dense visualization produced by interpolating sparse LK vectors (color-coded). (Right) Speed proxy (median pixel displacement $\times$ FPS) over sampled frames.

Figure 2: Interpolated mid-frames synthesized using forward/backward LK and blending. Small motions are reconstructed reasonably; larger displacements may produce ghosting.

Figure 3: Representative frame from the stabilized video (global translation compensated). The stabilisation reduces visible handheld jitter; residual parallax remains.

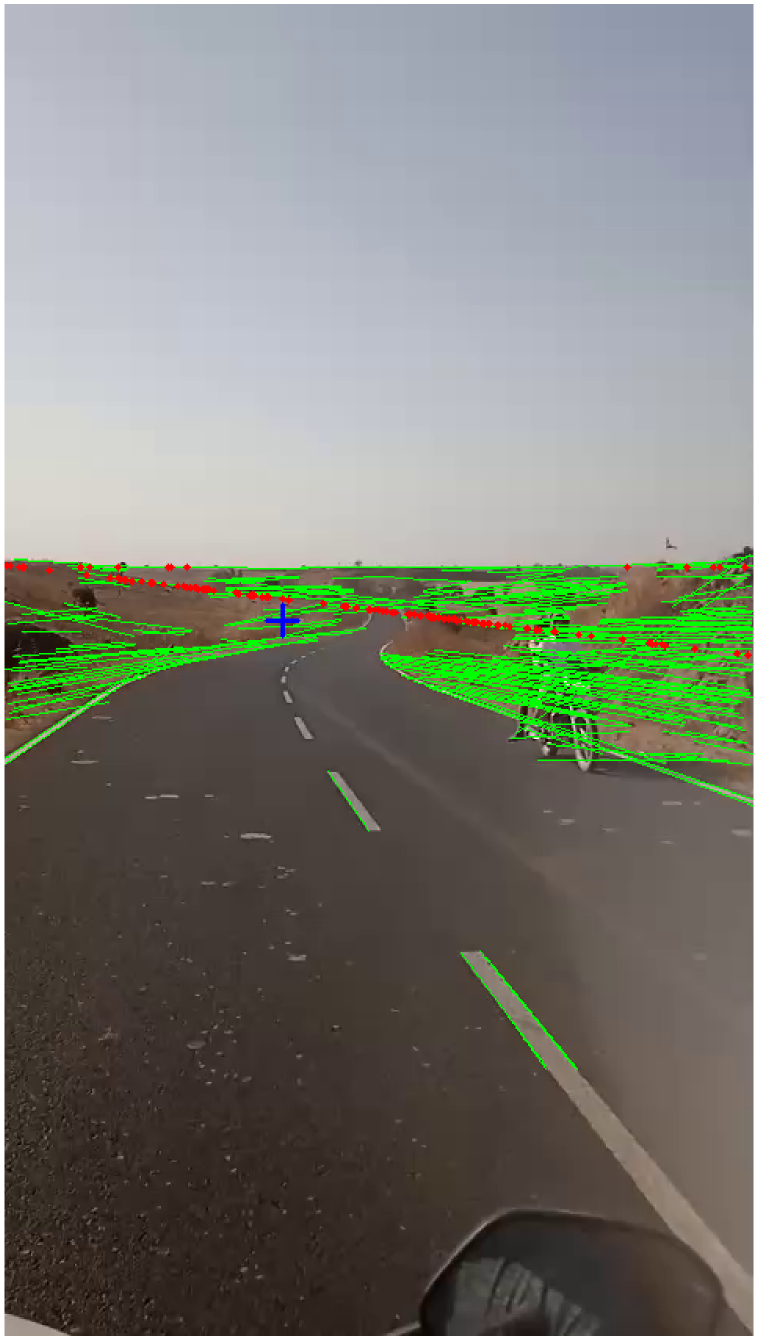

Figure 4: Vanishing-point estimation overlay: detected lines (green), sampled intersections (red), and estimated vanishing point (blue cross).

4 Discussion

The from-scratch Lucas–Kanade implementation provides a simple and fast sparse motion estimator useful for downstream tasks (speed proxy, global motion estimation). The sparse-to-dense interpolation approach used here is crude but sufficient for synthesizing plausible mid-frames when motion is small; for larger motions or occlusions, more advanced flow interpolation (or learned models) are preferable.

Stabilisation via median translation is effective for reducing shake but cannot handle rotation or parallax; an improved pipeline would estimate an affine or homography transform from robust matches, or use feature tracking and bundle adjustment.

Vanishing point detection is reliable in scenes with strong linear geometry (roads, railways) but degrades when lines are few or noisy.

5 Conclusions

We delivered a compact, self-contained implementation of core assignment tasks, produced qualitative outputs for each question, and packaged results and a short report in the repository. For final submission, the student should add their own short phone videos (privacy-compliant) and refine parameters.

Acknowledgements

Code and figures were generated using the scripts in this repository.

References

[1] Bruce D. Lucas and Takeo Kanade. An Iterative Image Registration Technique with an Application to Stereo Vision. 1981.